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Time-dependent Quantum Dynamics

Mathematics of Quantum Mechanics
MSc in Mathematical Engineering

Aronne Moltrasio

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Given the postulate on dynamics, we want to find an explicit expression of $\psi(t)$.

$$\psi(t) = U(t)\psi(s)$$

with s "starting time". Postulate require Schroedinger Equation

$$\begin{cases} i \hbar \frac{\partial}{\partial t} \psi(t) = -H(t)\psi(t) \\ \psi(0) = \psi_0 \end{cases} \quad (1)$$

Dependence on spatial coordinates is neglected.

"From A we construct $U(t) \equiv e^{itA}$ "

Theorem

Let A self adjoint operator, let $U(t) = e^{iAt}$, then

1. $U(t)_{t \in \mathbb{R}}$ is a strongly continuous unitary group.
2. $\psi \in D(A) \iff \exists$

$$\lim_{t \rightarrow 0} \frac{e^{iAt}\psi - \psi}{t} = iA\psi$$

"Any $U(t)$ is characterized by a unique A satisfying $U(t) = e^{itA}$ "

Theorem (Stone)

Let $\{U(t)\}_{t \in \mathbb{R}}$ be a strongly continuous one-parameter unitary group on a Hilbert space $\mathcal{H}$, then there exist a unique self-adjoint operator on $\mathcal{H}$ such that $U(t) = e^{itA}$.

Moreover, $\psi \in D(A) \Leftrightarrow \exists \lim_{t \rightarrow 0} \frac{U(t)\psi - \psi}{t} (= iA\psi)$

Proof (Stone's Theorem)

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Let $f \in C_0^\infty(\mathbb{R})$, $\varphi \in \mathcal{H}$

$$\varphi_f := \int_{-\infty}^{\infty} f(t)U(t)\varphi dt$$

Define $\mathbf{D} := \{\text{finite linear combinations of } \varphi_f\}$.

Recall: Approximate identity $j_\epsilon : [-\epsilon, +\epsilon] \rightarrow \mathbb{R}$ with $j_\epsilon * f \rightarrow f$ as $\epsilon \rightarrow 0$.

(Show that) **$\mathbf{D}$ is dense in $\mathcal{H}$:**

$$\|\varphi_f - \varphi\| = \left\| \int_{-\infty}^{\infty} j_\epsilon(t)(U(t)\varphi - \varphi)dt \right\| \quad (2)$$

$$\leq \int_{-\infty}^{\infty} \|j_\epsilon(t)\| \| (U(t)\varphi - \varphi) \| dt \quad (3)$$

$$\leq \sup_{t \in (-\epsilon, \epsilon)} \| (U(t)\varphi - \varphi) \| \int_{-\infty}^{\infty} j_\epsilon(t) dt \quad (4)$$

Define A: let $\varphi_f \in D$

$$A\varphi_f := \lim_{s \rightarrow 0} \left(\frac{U(s) - I}{s} \right) \varphi_f = \frac{1}{i} \varphi_{-f'}$$

The property $A\varphi_f = \frac{1}{i} \varphi_{-f'}$ can be computed.

Notice that $U(t): D \rightarrow D$, $A: D \rightarrow D$, $U(t)A\varphi_f = AU(t)\varphi_f$.

A symmetric:

$$(A\psi_f, \phi_g) = (\lim_{s \rightarrow 0} (\frac{U(s) - I}{is}) \psi_g, \phi) \quad (5)$$

$$= (\psi_f, \lim_{s \rightarrow 0} (-\frac{U(s) - I}{is}) \phi_g) \quad (6)$$

$$= (\psi_f, \frac{1}{i} \phi_{-g'}) \quad (7)$$

$$= (\psi_f, A\phi_g) \quad (8)$$

A essentially self-adjoint:

Assume $\exists u \in D(A^*)$ s.t. $A^*u - iu = 0$, then $\forall \varphi \in D(A) = D$. Then by computing

$$\frac{d}{dt}(U(t)\varphi, u) = (\lim_{t \rightarrow 0} \frac{U(t) - I}{t} \varphi, u) \quad (9)$$

$$= (iAU(t)\varphi, u) \quad (10)$$

$$= -i(U(t)\varphi, A^*u) \quad (11)$$

$$= -i(U(t)\varphi, iu) \quad (12)$$

$$= (U(t)\varphi, u) \quad (13)$$

one get the differential equation with solution (next slide)

$$f(t) = (U(t)\varphi, u) = f(0)e^t$$

but "U(t) bounded" causes an absurdity. Thus $\nexists u$ s.t. $A^*u = iu$, that is

$$\text{Ker}(A^* - i) = \{0\} \quad \text{and} \quad \text{Ker}(A^* + i) = \{0\}$$

by criterion for self-adjointness and corollary, A is e.s.a on D.

$$U(t) = e^{it\bar{A}} =: V(t):$$

By precedent theorem

$$\forall \psi \in D(\bar{A}) \quad (e^{it\bar{A}})' \psi = i\bar{A}\psi$$

Let $w(t)\psi := U(t)\psi - V(t)\psi$

$w(t)$ is strongly differentiable vector valued function.

$$w'(t) = iAU(t)\psi - i\bar{A}V(t)\psi = i\bar{A}w(t) \quad (14)$$

$$\frac{d\|w(t)\|^2}{dt} = \left(\frac{d}{dt}w(t), w(t)\right) + \left(w(t), \frac{d}{dt}w(t)\right) \quad (15)$$

$$= -i(\bar{A}w(t), w(t)) + i(\bar{A}w(t), w(t)) = 0 \quad (16)$$

Therefore $w(t)=0$ because $w(0)=0$. $\Rightarrow U(t)\psi = V(t)\psi \quad \forall t \in \mathbb{R}$,
 $\forall \psi \in D \Rightarrow U(t)\psi = V(t)\psi \quad \forall \psi \in \mathcal{H}$ because D is dense.



One can ask for less hypotheses:

1. $U(t)$ unitary and weakly continuous $\Rightarrow$ strongly continuous.

$$\|U\psi - \psi\|^2 = \|U\psi\|^2 + \|\psi\|^2 - (U\psi, \psi) - (\psi, U\psi) \quad (17)$$

$$\xrightarrow[t \rightarrow 0]{} 2\|\psi\|^2 - 2\|\psi\|^2 \quad (18)$$

$$= 0 \quad (19)$$

2. (**Von Neumann's Theorem**) Let $U(t)$ be a one-parameter unitary group on a separable Hilbert space $\mathcal{H}$. If $\forall \psi, \phi \in \mathcal{H}$, $(U(t)\psi, \phi)$ is measurable, then $U(t)$ is strongly continuous.

Definition

Given a strongly continuous one-parameter unitary group $U(t)_{t \in \mathbb{R}}$, the **infinitesimal generator** is that (unique) self-adjoint operator A for which $U(t) = e^{itA}$.

The Hamiltonian H is the infinitesimal generator of the unitary time-evolution.

Theorem

Let $\underline{t} \mapsto U(t_1, \dots, t_n)$ be a strongly continuous map of $\mathbb{R}^n$ into the unitary operators on separable $\mathcal{H}$, satisfying

$$U(0) = I \quad \text{and} \quad U(t+s) = U(t)U(s)$$

Then, $\exists A_j$ family of infinitesimal generators for the one-parameter group $U(0, \dots, 0, t_j, 0, \dots, 0)$ with domain of essential self adjointness D

$$D := \{\text{finite linear combination of } \varphi_f\} \quad \varphi_f := \int_{-\infty}^{\infty} f(t)U(t)\varphi dt$$

with $f \in C_0^\infty(\mathbb{R}^n)$, $\varphi \in \mathcal{H}$.

Theorem

Moreover, A_j commutes with any other A_i and $\exists$ a projection valued operator P_Ω on $\mathbb{R}^n$ such that

$$(\psi, U(t)\phi) = \int e^{it\lambda} d(\psi, P_\lambda \phi)$$

$\forall \psi, \phi \in \mathcal{H}$.

Significance: evolution along multiple commuting directions.

The aim is to look for unitary propagators $U(t,s)$ that satisfy

$$\frac{d}{dt}\varphi(t) = -iH(t)\varphi(t)$$

with initial condition $\varphi(s) = \varphi_s$.

Definition

A two-parameter family of unitary operators $U(t,s)$, with $t,s \in \mathbb{R}$, is called **unitary propagator** if it satisfies:

1. $U(t,t) = I$
2. $U(t,r)U(r,s) = U(t,s)$
3. $U(t,s)$ jointly strongly continuous in s and in t .

Particular case: H bounded.

Theorem (Dyson Expansion)

Let $t \mapsto H(t)$ be a strongly continuous map into **bounded** self-adjoint operators on $\mathcal{H}$. Then $\exists$ unitary propagator such that

$$\varphi_s(t) = U(t, s)\varphi_s$$

with $\varphi_s(s) = \varphi_s$, satisfies

$$\frac{d}{dt}\varphi_s(t) = -iH(t)\varphi_s(t)$$

where $\forall \varphi_s \in \mathcal{H}$.

Theorem

In particular the Dyson expansion

$$U_n(t, s) \varphi := \varphi + \sum_{n=1}^{\infty} (-i)^n \int_s^t dt_1 \int_s^{t_1} dt_2 \cdots \int_s^{t_{n-1}} dt_n H(t_1) \cdots H(t_n) \varphi,$$

converges with respect to the operator norm.

Definition and strongly continuity

$$U(t, s)\psi := \sum_{n=0}^{\infty} (-i)^n \int_s^t \int_s^{t_1} \dots \int_s^{t_{n-1}} H(t_1) \dots H(t_n) \psi \, dt_1 \dots dt_n$$

Each n^{th} term can be bounded by

$$(-i)^n \int_s^t \int_s^{t_1} \dots \int_s^{t_{n-1}} H(t_1) \dots H(t_n) \psi \, dt_1 \dots dt_n \leq \frac{|t-s|^n}{n!} \left(\sup_{t \in [s, t]} \|H(t)\| \right)^n \|\psi\|$$

Therefore, the series converges uniformly in the operator norm.

Therefore, strongly continuity is preserved and $U(t, s)$ is jointly strongly continuous.

$U(t,t)=I$ trivial

$$U(t,r)U(r,s) = U(t,s)$$

$$U(t,r)U(r,s)\psi = \left[\sum_{n=0}^{\infty} (-i)^n \int_r^t \int_r^{t_1} \dots \int_r^{t_n} H(t_1) \dots H(t_n) \psi \, dt_1 \dots dt_n \right] \\ \times \left[\sum_{k=0}^{\infty} (-i)^k \int_s^r \int_s^{r_1} \dots \int_s^{r_n} H(r_1) \dots H(r_n) \psi \, dr_1 \dots dr_n \right]$$

Where time ordering $s \leq r_1 \leq \dots \leq r_n \leq r \leq t_1 \leq \dots \leq t_n \leq t$ is managed by time ordered n-simplex

$$\Delta_n(r, t) = \{(u_1, \dots, u_n) : r \leq u_n \leq \dots \leq u_1 \leq t\}.$$

$$U(t, r)U(r, s)\psi = \sum_{n,k=0}^{\infty} (-i)^{n+k} \int_r^t \int_s^r \int_{\Delta_{n-1}(r,t) \times \Delta_{k-1}(s,r)} [H(r_1) \dots H(r_k) \\ \times H(t_1) \dots H(t_n) \psi \, dr_1 \dots dr_k dt_1 \dots dt_n]$$

Thanks to Fubini theorem and through the change of variables $(r_1, \dots, r_k, t_1, \dots, t_n) \rightarrow (\tau_1, \dots, \tau_k, \tau_{k+1}, \dots, \tau_{k+n})$. The result is

$$U(t, r)U(r, s) = U(t, s).$$

$U(t,s)$ is unitary, that is $U(t,s)^{-1} = U(t,s)^*$

We proceed through computations, first let's explicit each n^{th} component

$$[U(t,s)]_n = \left(\frac{-i}{\hbar}\right)^n \int_s^t dt_1 \int_s^{t_1} dt_2 \cdots \int_s^{t_{n-1}} dt_n H(t_1) \cdots H(t_n).$$

$$[U(t,s)^*]_n = \left(\frac{i}{\hbar}\right)^n \int_s^t dt_1 \int_s^{t_1} dt_2 \cdots \int_s^{t_{n-1}} dt_n H(t_n) \cdots H(t_1).$$

While

$$[U(s,t)]_n = \left(\frac{-i}{\hbar}\right)^n \int_t^s d\tau_1 \int_t^{\tau_1} d\tau_2 \cdots \int_t^{\tau_{n-1}} d\tau_n H(\tau_1) \cdots H(\tau_n).$$

Then one operates the change of variables

$$\sigma_k = t + s - \tau_k, \quad k = 1, \dots, n.$$

Then

$$d\tau_k = -d\sigma_k,$$

hence

$$d\tau_1 \cdots d\tau_n = (-1)^n d\sigma_1 \cdots d\sigma_n.$$

The inequalities

$$t \geq \tau_1 \geq \tau_2 \geq \cdots \geq \tau_n \geq s$$

are equivalent to

$$s \leq \sigma_1 \leq \sigma_2 \leq \cdots \leq \sigma_n \leq t.$$

Therefore one obtains

$$[U(s, t)]_n = \left(\frac{i}{\hbar}\right)^n \int_s^t d\sigma_1 \int_s^{\sigma_1} d\sigma_2 \cdots \int_s^{\sigma_{n-1}} d\sigma_n H(\sigma_n) \cdots H(\sigma_1),$$

and hence

$$U(s, t) = U(t, s)^*$$



Let ψ_k, ψ_l be the eigenfunctions of an Hamiltonian H_0 , and let λ_k, λ_l be the corresponding eigenvalues.

Let the system start in $\psi(0) = \psi_k$. If $\mathbf{H}(\mathbf{t}) = \mathbf{H}_0$ the state will remain unchanged.

$$\psi(t) = e^{itH_0}\psi_k = e^{it\lambda_k}\psi_k$$

While if $\mathbf{H}(\mathbf{t}) = \mathbf{H}_0 + \mathbf{V}(\mathbf{t})$ with $V(t)$ strongly continuous, bounded and self-adjoint but H_0 possibly unbounded, we deal with the *interaction representation* as follows.

Define $\tilde{V}(t) = e^{-itH_0} V(t) e^{itH_0}$, notice self-adjoint, bounded and strongly continuous. $\Rightarrow \exists \tilde{U}(t, s)$ corresponding propagator.

Set $U(t, s) = e^{-itH_0} \tilde{U}(t, s) e^{isH_0}$, then

$$\frac{d}{dt} U(t, s) \psi = -i(H_0 + V(t)) U(t, s) \psi$$

So, $U(t, s)$ is the corresponding propagator for $H_0 + V(t)$.

The probability of transition after a time t is $|(\psi_I, e^{-itH_0} \tilde{U}(t, 0) \psi_k)|^2$

By Dyson expansion

$$(\psi_I, e^{-itH_0} \tilde{U}(t, 0) \psi_k) = (\psi_I, e^{-itH_0} \psi_k) - i \int_0^t (\psi_I, e^{-it_1 H_0} \tilde{V}(t_1) \psi_k) dt_1 + \dots$$

$H(t) = iA(t)$ possibly unbounded

Theorem

Let X be a Banach space and let I be an open interval in $\mathbb{R}$. For each $t \in I$, let $A(t)$ be the generator of a contraction semigroup on X with $0 \in \rho(A(t))$, and define $C(t, s) := A(t)A^{-1}(s) - I$ so that

- (a) The $A(t)$ have a common domain D
- (b) For each $\varphi \in X$, $(t - s)^{-1}C(t, s)\varphi$ is uniformly strongly continuous and uniformly bounded in s and t for $t \neq s$ lying in any fixed compact subinterval of I .
- (c) For each $\varphi \in X$, $C(t)\varphi \equiv \lim_{s \uparrow t} (t - s)^{-1}C(t, s)\varphi$ exists uniformly for t in each compact subinterval and $C(t)$ is bounded and strongly continuous in t .

Theorem

Then for all $s \leq t$ in any compact subinterval of I and any $\varphi \in X$,

$$U(t, s)\varphi = \lim_{k \rightarrow \infty} U_k(t, s)\varphi$$

exists uniformly in s and t . Further, if $\psi \in D$, then $\varphi_s(t) \equiv U(t, s)\psi$ is in D for all t and satisfies

$$\frac{d}{dt}\varphi_s(t) = -A(t)\varphi_s(t), \quad \varphi_s(s) = \psi,$$

and $\|\varphi_s(t)\| \leq \|\psi\|$ for all $t \geq s$.

$$H(t) = H_0 + V(t) \text{ (interaction picture)}$$

with $H_0 = -\Delta$ on $L^2(\mathbb{R}^3)$, $t \mapsto V_1(t)$ and $t \mapsto V_2(t)$ continuously differentiable, respectively $L^2(\mathbb{R}^3)$ -valued and $L^\infty(\mathbb{R}^3)$ -valued.

$$V(t) = V_1(t) + V_2(t).$$

Consider $t \in [-T, T]$.

- $H_0 = -\Delta$ self-adjoint on $D(-\Delta) = H^2(\mathbb{R}^3)$.
- $V(t)$ bounded, symmetric, $t \in [-T, T]$. By Kato–Rellich

$(H_0 + V(t))$ self-adjoint on $D(-\Delta)$,

- Since $V(t)$ is uniformly bounded, one can define D s.t.
 $(H_0 + V(t) + D) \geq \frac{1}{2}$.
- Thus $i(H_0 + V(t) + D)$ and $-i(H_0 + V(t) + D)$ generate contraction semigroups (for each t) and $(\pm i(H_0 + V(t) + D))^{-1}$ exists for $t \in [-T, T]$.

- Hypotheses on $t \mapsto V_1(t)$ and $t \mapsto V_2(t)$ imply that $i(H_0 + V(t) + D)$ and $-i(H_0 + V(t) + D)$ satisfy the hypotheses (b) and (c) of previous theorem.
 $\Rightarrow \exists U^+(t, s)$ and $U^-(t, s)$. They satisfy

$$\frac{d}{dt} U^\pm(t, s) \psi = -(\pm i(H(t) + D) \psi$$

- Now define

$$\tilde{U}(t, s) = \begin{cases} U^+(t, s), & s \leq t, \\ U^-(s, t), & t \leq s, \end{cases}$$

and

$$U(t, s) = e^{iD(t-s)} \tilde{U}(t, s).$$

→ we got as result a unitary propagator satisfying the Time-dependent Schroedinger Equation. Let $\psi \in \mathcal{D}(H_0)$

$$\frac{d}{dt} U(t, s) \psi = -iH(t)U(t, s)\psi$$

Thank you!

1. M. Reed, B. Simon, Methods of Modern Mathematical Physics, vol. 1: Functional Analysis, Elsevier Science, 2012, ISBN 978-0-125-85050-6.
2. M. Reed, B. Simon, Methods of Modern Mathematical Physics, vol. 2: Fourier Analysis, Self- adjointness, Elsevier Science, 2012, ISBN 978-0-125-85002-5

Where the interval $[s, t]$ is divided in k subintervals to define the approximate propagator

$$U_k(t, s) := e^{-(t-s)A(\frac{j-1}{k})} \text{ if } s, t \in [\frac{j-1}{k}, \frac{j}{k}]$$

$$U_k(t, s) := U_k(t, r)U_k(r, s) \text{ otherwise, iteratively.}$$

